

24/7

HW

8) what are the various problems which you have observed in community & society. pick any 5 problems, find out if they can be solved using Data Structures & Algorithms.

Ans

1) Traffic Jams

Roads and junctions = a graph. Shortest path algorithms find the fastest route & reduce traffic.

2) Long Queues (Hospital, bank, ration shops)

This can be simply solved by a Queue - first come, first served.

3) Garbage collection not covering all areas properly

Treat houses/streets as points on a graph. Find the shortest path covering all points so the truck doesn't waste fuel.

4) Duplicate entries in voter ID lists or ration cards.

Using Hashing, we can quickly detect duplicate entries & clean up records instead of checking manually one by one.

5) Managing student / patient records ne-
ly (add, search, delete easily).

Managed by using a tree or list to
store records neatly so they're easy
to search, add, or update.